Product Requirements Document (PRD) — WariMitra

Document Identifier: WM-PRD-2026-V2.0

Product Name: WariMitra ("Digital Wari Ecosystem")

Product Category: GovTech / Smart Mass Gathering & Public Safety Management Platform

Product Owner / Lead BA: Senior Product Analyst & Solution Architect

Release Target: Version 2.0 (Annual Pandharpur Pilgrimage Cutover)

Repository Context: `c:\Users\mustafa\Desktop\WariMitra`

1. Product Vision & Value Proposition

1.1 Product Vision Statement

"To establish WariMitra as India's premier digital public infrastructure for mass gatherings, transforming the 700-year-old Pandharpur Wari into a safe, connected, and dignified journey where every pilgrim is protected by sub-second emergency response, proactive crowd intelligence, and transparent public governance."

1.2 Core Value Pillars

1. **Infallible Pilgrim Safety:** Sub-2-second panic dispatch with zero-text physical gesture triggers.
2. **Operational Decoupling:** Allowing healthcare, police, municipal sanitation, and temple trust teams to operate independently while synchronizing on a single shared operational picture.
3. **Radical Demographic Inclusivity:** Designing specifically for low-end Android smartphones, rural networks, and Marathi-speaking elderly users.
4. **Verifiable Governance:** Anchoring high-priority public safety records to tamper-evident public cryptographic ledgers at ₹0 recurring infrastructure cost.

2. Target Personas & User Archetypes

Persona 1: Tukaram Patil (The Rural Pilgrim)

- **Age & Background:** 68 years old, sugarcane farmer from Satara district, walking his 12th Wari.
- **Device & Tech Literacy:** Entry-level 4G Android smartphone; struggles with small fonts and complex

- **Device & Tech Literacy:** Entry-level 4G Android smartphone; struggles with small fonts and complex menus; reads Marathi fluently.
- **Key Needs:** Easily finding drinking water tankers, clean toilets, avoiding stampedes, and knowing when to enter temple queues.
- **Quote:** "*When my feet hurt or if I get separated from my village group in the evening crowd, I need help immediately without getting confused by English apps.*"

Persona 2: Inspector Vaishali Shinde (The Police Dispatcher)

- **Age & Background:** 38 years old, Maharashtra State Police officer stationed at Saswad Mobile Command Center.
- **Tech Literacy:** High; proficient with web applications, GIS maps, and VHF radio dispatch.
- **Key Needs:** Filtering false alarms, receiving precise GPS locations for emergency calls, monitoring crowd density heatmaps along highway bottlenecks.
- **Quote:** "*Radio descriptions like 'near the yellow tent' waste 30 minutes of our response team's time. Give me a pinpoint on a map with an ETA.*"

Persona 3: Dr. Ramesh Kadam (Medical Camp Officer)

- **Age & Background:** 45 years old, District Civil Surgeon overseeing a 20-bed emergency field clinic in Lonand.
- **Tech Literacy:** Moderate; uses hospital management software and tablets.
- **Key Needs:** Directing incoming ambulances away from already full camps, tracking bed occupancy in neighboring sectors, stocking adequate antivenom.
- **Quote:** "*We cannot have 50 heatstroke patients arriving at our clinic when the neighboring relief camp 2 km down the highway has 15 open beds.*"

3. Product Goals & Non-Goals

3.1 Product Goals

- **G-01:** Deliver distress notification from mobile screen tap to command center display in ≤ 2.0 seconds.
- **G-02:** Achieve 100% localization in Marathi for all pilgrim-facing mobile screens.
- **G-03:** Provide proximity-sorted medical camp queries with < 500 ms response time.

- **G-04:** Provide non-repudiable audit logs of all SOS incidents via zero-cost Web3 cryptographic anchoring.

3.2 Product Non-Goals (Explicit Boundaries)

- **NG-01:** WariMitra is **not** a social media network or generic chat application; family messaging is restricted to administrative alerts and group location telemetry.
- **NG-02:** WariMitra does **not** process monetary transactions, temple donations, or commercial vendor payments.
- **NG-03:** WariMitra does **not** replace physical police presence or official 108 emergency telephony; it augments them with digital coordinates.

4. MoSCoW Feature Prioritization & Rationale

MUST HAVE	SHOULD HAVE
• 3-Second Hold SOS Button	• Ad-hoc Family Group Tracking
• Idempotency & Rate Limiting	• 30-Second Location & Battery Ping
• Proximity Medical Camp Radar	• Web3 0-ETH Audit Proof Anchoring
• Virtualized Command Incident Feed	• Crowdsourced Waste/Sanitation Log
• Native Marathi Localization	• Digital Darshan Slot Booking Pass
COULD HAVE	**WON'T HAVE**
• Temple Queue Little's Law Calc	• Automated AI Facial Recognition
• NOAA Heatstroke Risk Calculation	• Deep Learning Crowd Surge Vision
• Cultural Saint & Abhang Library	• In-App Monetary Temple Donations
• Municipal Bio-Toilet Clean Status	• Offline SMS/IVR Voice Gateway

Detailed Rationale for Classification:

1. **Must Have:** Features directly tied to life safety, distress dispatch, and avoiding catastrophic stampedes. If the 3-second hold SOS or proximity medical radar fails, the platform fails its primary public mandate.
2. **Should Have:** High-value operational enhancements (family tracking circles, Web3 audit trails, sanitation logs) that significantly elevate user experience and governance accountability.
3. **Could Have:** Value-added cultural content (saints, devotional songs) and heuristic calculations that enhance user engagement without impacting core triage operations.

4. **Won't Have (v2.0):** Complex machine learning (facial recognition, deep video analytics) and telephony infrastructure (IVR/SMS) that would introduce prohibitive cloud hosting costs, technical debt, and privacy/DPDP compliance complexities.

5. High-Level Release Strategy & Phasing

- **Phase 1 (Alpha / Emulator Validation):** Local validation of Cloud Functions, Firestore Security Rules, and Flutter UI using Firebase Local Emulator Suite (`ARCHITECTURE.md:512`).
- **Phase 2 (Beta / Staging Pilot):** Deployed to Firebase staging project in `asia-south1`; field simulation with 200 volunteer devices in Pune district.
- **Phase 3 (Production Cutover):** Deployed to production Blaze plan; live operational rollout across the 21-day pilgrimage corridor (`ARCHITECTURE.md:525-535`).
- **Post-Event Review:** Evaluation of incident audit logs, serverless execution costs, and system performance benchmarks.